Progressive gate-unfreezing of the QFT operator

One gate thawed per stage, trained to a plateau — three unfreeze orderings × two initialisations on DIV2K-8q

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Question

The `QFTBasis(8, 8)` (72 U(2) gates) starts either at the QFT-family **identity** ($T_{t=0}(x) = x$) or **Haar-random**. We **unfreeze one gate per stage**, train it to a plateau, then thaw the next. Does the **order** — block-growth (`bg`), left→right (`lr`), right→left (`rl`) — or the **initialisation** change the trajectory or the endpoint?

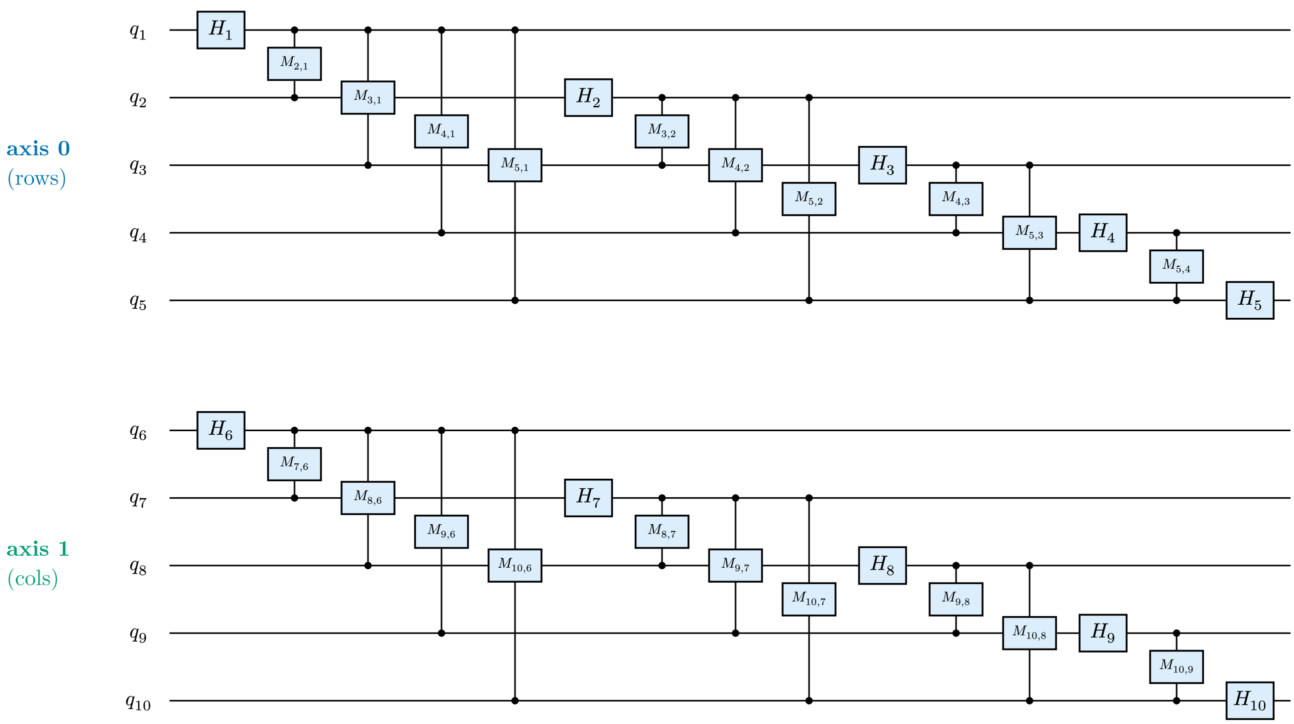

Figure 1: The **2-D** QFT$(5, 5)$ — two independent QFT(5) blocks, one per image axis (**axis 0** $= q_1..q_5$, rows; **axis 1** $= q_6..q_{10}$, columns). H_i = Hadamard on q_i; $M_{j,i}$ = controlled-phase between q_j and q_i $(j > i)$. The DIV2K runs use the analogous QFT$(8, 8)$ (axis 1 $= q_9..q_{16}$, 72 gates).

Every ordering thaws the gates of Figure 1 one at a time, on **both** axes:

- **block-growth** (`bg`): grow a **square $2^k \times 2^k$ block** — block-stage k thaws qubit k on both axes. H_1, H_6; then $H_2, M_{2,1}, H_7, M_{7,6}$; then $H_3, M_{3,1}, M_{3,2}, H_8, M_{8,6}, M_{8,7}$; … up to $k\{=\}5$ $(H_5, ..., M_{5,4}, H_{10}, ..., M_{10,9})$. Long-range couplings last. (At $m = 8$ the axis-1 partner of H_1 is H_9 — hence the early H_9 in the dynamics figure.)
- **left→right** (`lr`): QFT construction order, axis 0 then axis 1 — $H_1, M_{2,1}, ..., H_5$, then $H_6, M_{7,6}, ..., H_{10}$.
- **right→left** (`rl`): the reverse — H_{10} first, H_1 last.

Setup

Fixed batch of 50 training images; `MSELoss` on the top-10% coefficients. pdft's JIT Adam step runs with a per-stage frozen set (moments reset per stage); a Riemannian grad-norm probe gives the plateau signal. A stage ends at $\|g\| < 10^{-5}$ **or** $|\Delta L| < 10^{-5}$ (min 5 steps), else an 800-step cap. Random init: Haar $U(2)$ per Hadamard, uniform phase per controlled-phase, seed shared across orderings.

Training dynamics

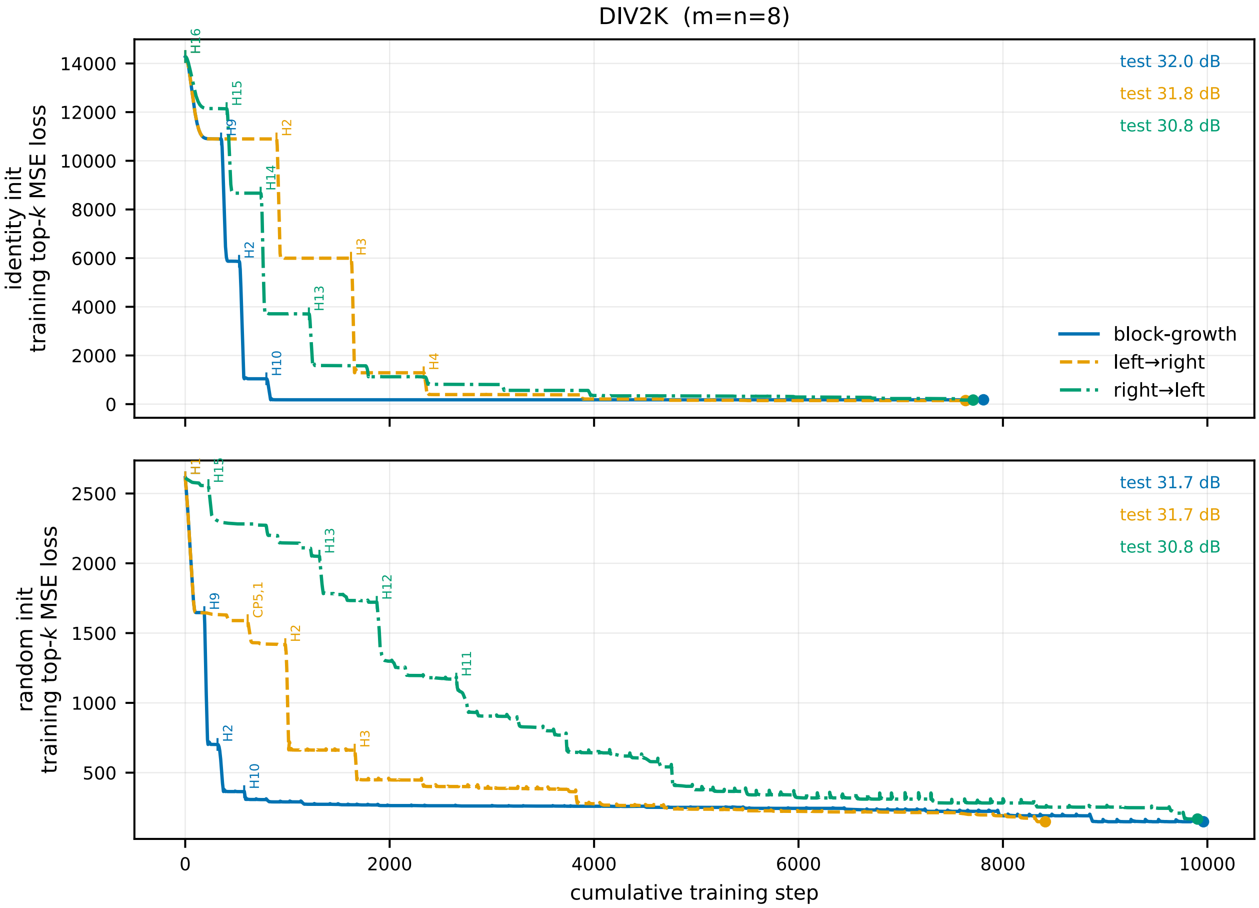

Figure 2: Absolute **training** top-k MSE loss vs cumulative step, one curve per ordering (rows: identity / random init). Tick labels mark **which gate** (e.g. `H7`, `CP3,1`) was thawed at each of the largest drops. The y-axis is the **training** top-10%-coefficient MSE loss; endpoints are annotated with the separate **test PSNR@ ρ{=}.20** (image reconstruction on held-out data) — the two are different metrics, so the curve's height does not determine the PSNR. Per-cell loss+grad-norm staircases: `div2k_8q/<init>/figures/`.

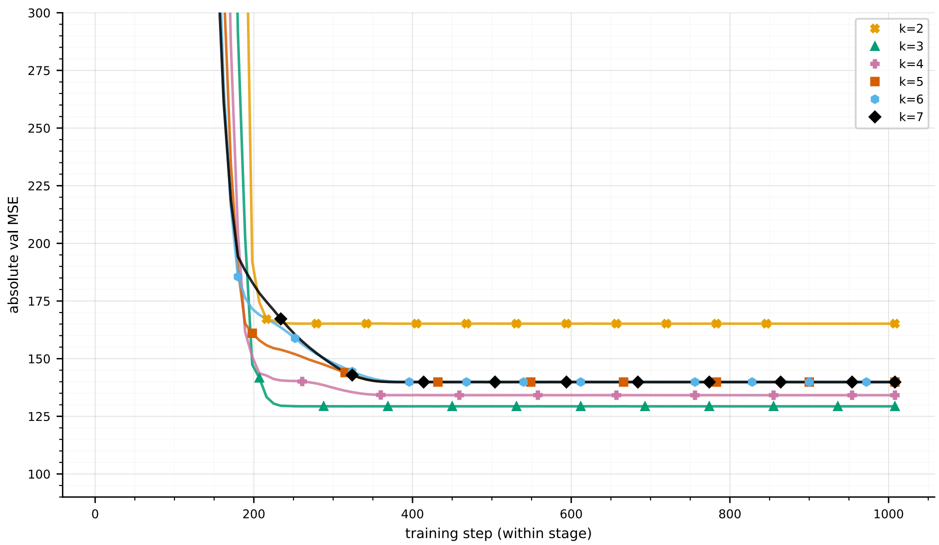

Figure 3: **Block-size** curriculum (`qft_progressive`, DIV2K) for comparison: per-stage validation MSE for block sizes $k = 2..7$, each one whole $2^k \times 2^k$ QFT trained 1008 steps. Eight whole-block stages here vs the 72 one-gate-at-a-time stages above — same ≈31.7 dB endpoint (table below).

Reconstruction PSNR (DIV2K)

Gate-unfreeze endpoints **together with** the `qft_progressive` block-size baseline (train all $k(k+1)$ gates of a $2^k \times 2^k$ QFT at once, per stage k). DIV2K test PSNR (dB) at four keep ratios ρ:

configuration	ρ{=}.05	ρ{=}.10	ρ{=}.15	ρ{=}.20	**train MSE**
unfreeze · identity · bg	19.4	26.9	29.7	**32**	178.2
unfreeze · identity · lr	25.2	**27.9**	**30**	31.8	146.6
unfreeze · identity · rl	24.8	27.2	29.1	30.8	167.8
unfreeze · random · bg	**25.2**	27.8	29.8	31.7	148.6
unfreeze · random · lr	**25.2**	27.8	29.8	31.7	148.6
unfreeze · random · rl	24.8	27.2	29.1	30.8	167.8
block-size · k=1 (2×2)	9.4	12.5	16.2	21.1	—
block-size · k=2 (4×4)	19.1	26.9	29.8	32	—
block-size · k=3 (8×8)	25.2	**28.1**	**30.3**	**32.3**	—
block-size · k=4 (16×16)	**25.3**	28	30.1	32	—
block-size · k=5 (32×32)	25.2	27.8	29.8	31.7	—
block-size · k=6 (64×64)	25.2	27.8	29.8	31.7	—
block-size · k=7 (128×128)	25.2	27.8	29.8	31.7	—
block-size · k=8 (256×256)	25.2	27.8	29.8	31.7	—
classical · block-DCT 8×8	**26.1**	**29.4**	**31.9**	**34**	—
classical · block-FFT 8×8	24.5	27.1	29.1	30.8	—

train MSE = final top-10%-coefficient training loss (absolute, train batch) — a different metric from the PSNR columns, shown per gate-unfreeze run; not defined for the block-size / classical references.

All learned QFT schemes reach the **same** QFT(8, 8) endpoint, ≈31.7 dB @ρ{=}.20: the block-size curriculum saturates by $k = 2$ and the gate-unfreeze sweep lands there from either init and any order — **schedule-independent** (`rl` trails ≈1 dB; every gate-unfreeze stage ended on the loss-Δ plateau, none at the step cap). For reference, the best **any** prior trained basis reaches is 33.7 dB (`rich/identity`). The classical **block-DCT 8×8** is in fact the strongest here (34 dB @ρ{=}.20), ahead of every learned QFT basis; **block-FFT 8×8** is weaker (30.8 dB) — the QFT-family operators sit between the two classical block transforms.

Is the fixed top-10% training objective a confound? The classical transforms are inherently rate-matched (keep top-ρ of a **fixed** operator at every ρ), while the learned QFT trains once at top-10% and is scored at 5–20%. This does not explain its deficit: (i) at the **matched** rate $\rho\{=\}.10$ the classical block-DCT (29.4 dB) still beats the best gate-unfreeze (27.9 dB); and (ii) retraining the QFT with top-k matched to each ρ raises the matched-rate PSNR by $\leq$ **0.07 dB** (controlled all-at-once probe, `reference/rate_matched_div2k.json`) — the operator's energy compaction is essentially rate-agnostic, so the top-k choice is not the issue.

Reading

The staircase shows each gate's marginal value: the big drops are the **early** gates (the marked Hadamards, e.g. `H9`, `H2` for block-growth); later gates only fine-tune. One caveat on the endpoint labels — the train-loss order need not match the test-PSNR order (`bg` has the **highest** train MSE yet the **best** test PSNR): the loss is a top-10%-coefficient proxy on the train batch, PSNR a 20%-keep reconstruction on the test set.